From Research Questions to Publication: Best Practices and AI Integration for Modern Scientific Writing

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ABSTRACT

Any research endeavor is not fully complete until it is properly communicated and published in a journal or conference. Publication in a journal that applies established peer review processes is a fundamental process of scientific communication. The decision to write, while making the effort to write well, is the cornerstone of scientifically literate individuals. Poor writing indicates poor thinking, and both are disastrous for research and the publication of research results. To write and publish a convincing scientific paper, many questions must be addressed, including what to write, where to publish, what style to apply, how to respond to a referee, and any additional questions that may arise during the process and even after publication.

In recent years, the advent of AI (*Artificial Intelligence*) driven tools has introduced new dynamics into the scientific writing process. While these tools can enhance writing quality and efficiency, they also pose ethical challenges both for authors and publishers alike, issues that must be navigated carefully. This manuscript provides basic guidelines, particularly for Early Career Researchers (ECR), that the author hopes will be useful in the effective and efficient writing and publication of scientific papers, including best practices for the responsible use of AI in scientific writing.

Keywords: Scientific Writing, Research, AI in Scientific Writing, Ethical Considerations.

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INTRODUCTION

Scholarly writing is an essential element of any academic endeavor and can be best characterized by a systematic approach to presenting ideas. It is widely utilized by scholars and educators to articulate data-driven arguments and logical analyses in scientific works. This style of writing facilitates comprehensive understanding of a subject, enabling authors to delve into concepts and develop well-substantiated theories, models or conclusions. Despite its importance,

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academic writing can be challenging, with the level of difficulty varying according to the writer's expertise and the field of study [1], [2], [3].

Scientific writing almost always begins with a question that cannot be definitively answered based on previous experience[3]. The list of questions is varied and unlimited, as are the complexities of the phenomena under study. When we seek beyond our own knowledge and experience to investigate, research, and use materials or methods beyond our own resources, we are conducting a research process. Research can be the study of a subject through first-hand experimental observations, a new mathematical proof, conducting a survey, analyzing data from a time series or the results of a computer simulation, or critically reviewing previous studies conducted by other researchers and presented in books, articles, or presentations. In most cases, the research process takes us through a combination of all the aforementioned activities.

The first significant written production that most Early Career Researchers produce is a thesis proposal, a progress report of their thesis work, or a short article written in conjunction with the thesis advisor or a tutor. In academia, this exercise is followed by the writing of the thesis or dissertation. To encourage early attention to the process required for scientific writing, most universities require a detailed prospectus, including a literature review, before granting permissions for the execution of research projects. This article does not aim to become a catalog of how to conduct research or write scientific articles. However, it aims to provide the reader with a generic overview of the scientific writing process, as well as some key points that can help coordinate such a process. Keep in mind that there is never a single “correct” way to approach a writing task; what is presented here is just but one approach and may not even be the best.

In recent years, AI-driven tools have emerged as valuable aids in scientific writing [4], [5], [6]. These tools can assist young researchers by offering suggestions for sentence structure, grammar, and style, ensuring that their manuscripts are clear and coherent. This is particularly important for those who are non-native to english. For instance, existing AI tools may assist non-native English speakers with translations into English with a desired tone and style. Such tools can also correct English grammar and typos. AI can also help in the organization of literature reviews, identifying key papers and summarizing relevant findings. By automating repetitive or time-consuming tasks, AI allows researchers to focus more on the creative and analytical aspects of their work, thereby enhancing the overall quality and efficiency of scientific writing.

At the time of writing the present document, some publishers and academic organizations continue debating on how AI can be addressed and suggest that in regards to the use of GPT (*Generative Pre-trained Transformers*) tools in scientific writing, it should be limited or not considered as part of the process[7], [8], [9], [10]. Although many questions are left open as to how an author can use AI tools, in particular LLMs (*Large Language Models*) for writing assistance particularly now

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that many commercial word processors are incorporating AI tools by default. The debate also continues about how precisely guidelines, policies and concepts of academic integrity should evolve in response to the increasing availability of generative AI tools, with no clear settled consensus yet. Many anticipate that norms surrounding incorporation and acknowledgment of LLMs generated text, images or code in scientific literature will change considerably in the near future[7], [11], [12].

However, the use of AI in scholarly writing also poses significant ethical challenges[13], [14], [15]. There is a risk of misuse, where dishonest researchers might use AI to generate content that violates ethical norms, such as plagiarism or fabricating data. AI-generated text can sometimes obscure the originality and authenticity of the research, leading to questions about the validity of the work. Therefore, it is crucial for researchers to use AI tools responsibly, ensuring transparency and integrity in their writing processes. Responsible acknowledgment and disclosure of the use of an AI-bot or tool in preparing specific sections of an article is absolutely necessary [16]. Table 1 summarizes the main domains in which AI is currently assisting scholarly writing and how such a process can take place:

Table 1. Domains in which AI can assist scientific writing

Domain	**How can AI assist**
Literature Review	AI tools can rapidly scan and analyze vast amounts of literature to identify relevant studies, summarize key findings, and highlight gaps in existing research.
Grammar and Style Improvement	AI-driven platforms like Grammarly and Hemingway Editor as well as as most open source LLMs can detect and correct grammatical errors, enhance readability, and suggest improvements in writing style.
Reference Management	Tools like EndNote, Mendeley and Zotero help in organizing and formatting citations and references according to specific journal guidelines, ensuring accuracy and consistency.
Plagiarism Detection	AI-powered plagiarism checkers like Turnitin, Dupli Checker and Copyscape can detect potential plagiarism by comparing the text against extensive databases of published works.
Data Analysis	AI can assist in analyzing complex datasets, identifying patterns, and generating visualizations, which can be included in the manuscript to support findings.
Journal Selection	AI tools can recommend suitable journals for manuscript submission based on the content, scope, and impact factors of the research.
Peer Review	AI can aid in the peer review process by suggesting potential reviewers, screening for conflicts of interest, and analyzing reviewer comments for common themes.
Proofreading and Editing	AI-driven tools can help in proofreading and editing manuscripts, ensuring that the final document is free of errors and meets the required standards.
Content Generation	AI can assist in generating content by providing initial outlines or drafts, summarizing sections, or suggesting relevant literature and research

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	questions.
Ethical Compliance	AI tools can screen manuscripts for ethical compliance, such as checking for conflicts of interest and adherence to ethical guidelines in research.

One of the primary difficulties for young early career researchers and new authors is crafting their study in a way that is both clear and engaging, effectively transforming their research findings into a persuasive narrative and their data into significant scientific contributions. Two pieces are fundamental for success. The first key component that provides a foundation of valuable research is a robust research design and meticulous execution. The second important part involves the way the findings are communicated—through both style and structure—, these two aspects play a crucial role in the study's acceptance, dissemination, and impact within the academic community and beyond[2].

1. Preliminary Questions to Ask About Research Design

Scientific research requires careful planning and consideration of various factors. Table 2 summarizes some essential questions to guide the design of any research endeavor, including considerations for the use of AI tools and the potential use of your collected data for training Machine Learning (ML) models by others:

Table 2. Some preliminary Questions to consider during research design

Question	**Examples of how to solve it**
Do I know what I am doing?	Have I outlined an action plan (a protocol, workflow plan or experimental design) for what I intend to do?
What are the real contributions of my research?	Am I addressing an important question? Am I aware of the key milestone works in this field? Have I identified the existing gap/challenges of this field?
Have I planned experiments that meet the required standards and/or protocols for publication?	Do the proposed experiments meet acceptable standards? Could the work I am doing adversely affect the environment? Are there any ethical considerations to consider while conducting my experiments? Should I adopt an already accepted experimental or computational protocol?
What practical, legal and/or political considerations need to be taken into account?	Could the publication of my results violate international treaties? Could the publication of results invalidate the application for a patent? Are there any legal issues if I publish my results in open access databases? Is there a Confidentiality Data Agreement that can be infringed by my publication?
How do I ensure my records are complete?	How do I register what I read? How will I keep a record of the work as it progresses? How do I ensure that my observations can be accessed in the future when needed by others or my research group? Is there a need for an external repository?
How do I ensure the long term	For certain types of unique or specialized data, there are

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care of the data I work with?	carefully-curated, web-integrated, dedicated repositories; which one, if any, should I upload my data to. How should I apply the FAIR guiding principles[17] for data management?

In the evolving landscape of academic and scientific research, the incorporation of AI into idea development and research design is becoming increasingly impactful. AI algorithms enhance the brainstorming process from the outset by offering insights based on current trends, historical data, and interdisciplinary studies, thereby significantly enriching the generation of ideas. Table 3 summarizes some questions about the use of AI in research design:

Table 3. Questions about the use of AI in research design

Question	**Examples of possible answers and solutions**
How can AI assist in my research design?	AI can help by analyzing large datasets to identify trends and patterns that may not be immediately evident. It can also assist in generating hypotheses based on existing data and literature. It may help with the identification of knowledge gaps.
What are the ethical considerations of using AI in my research?	Ensure that AI-generated content does not violate ethical norms, such as avoiding plagiarism and maintaining transparency about the use of AI-driven tools.
How will I ensure the validity and originality of AI-generated content?	Use AI as an aid, not a replacement for original thinking. Verify and critically evaluate AI-generated suggestions and data by contrasting with original sources whenever possible. Always keep in mind that AI bots are known to generate erroneous citations and/or data and even fabricate spurious information.
How can AI tools improve the efficiency of my research design?	AI can automate repetitive tasks, such as data collection and analysis. If properly used, AI can assist in identifying existing gaps in the knowledge that can be leveraged for more significant results. Integration with 3-D printing technologies and automated synthesis leads to rapid prototyping and development of new materials, structures or molecules for example.
Should I acknowledge the use of AI in my research design or my paper preparation?	As a good practice and keeping with the current trends, acknowledge in the Methods, Experimental or in Acknowledgments Sections, the use of any AI tools or AI-bots to prepare your manuscript. Clearly stating which parts of the manuscript use output from AI-bots and provide the prompts and questions used as supplemental information, if required.
Is the use of AI legal in the context of my research?	To train LLMs, the companies behind them draw on vast amounts of data and information extracted from the Internet and other sources much of which is copyright-protected material[18]. It is not entirely inconceivable that in the coming years some AI tools may actually be made illegal or be limited due to violations of intellectual property.

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Once this preliminary exercise on research design is completed, a somewhat more difficult question must be asked. Is the study I am currently conducting something really worth writing about? The answer to this question can only be yes if the results and conclusions meet one of the following requirements[1]:

- They are reasonably consistent, reproducible, and complete.
- They represent a significant experimental or theoretical extension of current knowledge.
- They represent a significant advance in the practical application of known principles.
- They push the frontier of knowledge one step forward.

In other words, are the results of your study really new, true, and significant? If the answer to all these questions is negative, it is better to postpone any publication effort. Sometimes a subject that originally seemed promising appears inappropriate in its current form or state of development due to for example new theories or methods. It is necessary then to postpone the publication effort until more thorough work has been done or new models or protocols have been adopted. A fundamental objective --difficult to follow in current times-- should be to produce a few publications of appropriate quality and impact rather than many mediocre ones. Producing low-quality publications or getting involved in misconduct practices due to pressures for publications can be problematic in the future of any researcher [19].

In recent years, several vexing problems have emerged within the scientific community, those related to paper fabrication networks or so call “paper mills”, ghostwriting companies and or groups that produce pseudo-scientific papers and sell authorship positions[20], [21]. A different problem is the so-called “citation cartels” or “citation networks”[22], [23] groups of unscrupulous researchers that with the purpose of increasing the number of citations for their work create citation cartels or networks, where members cite each other in order to increase their own number of citations or push the impact factor of certain dubious journals. A more recent problem generated with the arrival of AI is that of “soft plagiarism” and the so-called “tortured phrases” concept[24], these forms of plagiarism involve paraphrasing, --the change of original words by synonyms--, with the help of AI tools or online translators to evade plagiarism detection. In such a modality, authors copy from various sources, paraphrase the information, and paste such information to their own articles not paying attention to correct phrases or words that do not make any sense such us ‘*counterfeit consciousness’* in lieu of ‘*Artificial Intelligence*’ or ‘*colossal information*’ for ‘*big data*’ to cite some pointing examples[24]. Compilations of such phrases are available in the literature[25] and their numbers continue growing.

The “publish” or “perish” mantra that has permeated the academic world in recent decades, fueled by institutions and countries that provide financial incentives to reward authors who seek to publish in high index journals, has fostered and given way to fraud and misconduct with many documented cases reported[26]. Such misconduct by researchers has led to an increased number of retractions which continues on the rise, reaching all time records in 2023[27]. This problematic misconduct issue, sadly contributes to the ‘reproducibility crisis’ in various research

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areas[28],[29], which affects the trust people and funding organizations place in science and its results[30]. How we do science and, particularly, how effective and transparent we are in its communication, matters.

Once a question worth exploring has been identified, a literature review is almost always the first step in scientific research. The literature review reveals reports of similar work, helps to better focus the research question, and suggests appropriate routes for publication. AI can significantly enhance the efficiency and effectiveness of the literature review process. AI-driven tools can quickly scan vast amounts of academic papers, extracting key insights and summarizing relevant findings. These tools can help identify trends, gaps in the literature, and potential research questions by analyzing citation patterns and content. Additionally, AI can assist in organizing and managing references, ensuring that all relevant literature is accounted for and easily accessible. By automating these tasks, researchers can save valuable time and focus more on the critical analysis and synthesis of the information, ultimately leading to a more comprehensive and robust process of literature review.

This stage generally results in the management and analysis of a significant volume of articles, books, presentations, etc., which makes effective information organization an obligation, not an option, for a successful scientific writer. It is essential to have some system to organize all the information generated in a research project. What system is most effective? There is no satisfactory answer to such a question. Each situation requires a different style of organization to be carried out satisfactorily; what is certain is that most individuals perform better when they apply a technique associated with their own personal style and energy level. The important thing is that whatever system is used, it must be implemented promptly and followed consistently.

For example, AI's capability in text extraction and semantic analysis could prove to be vital for managing the growing data found in scientific literature in certain fields. AI tools, such as OpenAI's GPT, Google's Gemini, or Meta's Llama can analyze large datasets and produce high-quality content, though careful oversight is required to avoid generating misleading material or content. This ability to handle data is crucial for identifying trends, gaps, and emerging themes within each field.

2. The Message Determines the Medium

Once the literature review has been conducted and the research is in progress, it is necessary to give additional considerations to the writing task. At this point, a savvy writer asks themselves the following questions:

- What message am I trying to convey?
- What format is best for the message I want to convey?
- Who might be interested in my message?

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- Where should this article be published?
- How accessible it should be?

You should be able to answer these questions yourself, but sometimes it is appropriate to discuss these questions with more experienced colleagues. Most researchers suffer from the same difficulties when judging their own work; they tend to inflate the results and overestimate the size and nature of the potential audience.

Most of us are interested in gaining some recognition for the work we do. The way a study is presented, how and where it is published can greatly determine the degree of recognition and, in fact, whether the work receives any recognition at all. For a researcher to receive professional recognition for discovering something new, it is not enough to be the first to study it —the researcher must be the first to publish the information in a "valid" manner. A "valid" scientific publication has a series of essential components that distinguish it from any other type of publication:

- It is the first publication of research results.
- It is in a form in which other researchers in the field can reproduce and replicate the results and test the derived conclusions.
- It appears in a relevant journal or another documentary source.
- It is easily available to the scientific community.
- The article contains specific information.
- It is organized in a certain way, *i.e.*, it has a certain format.

A scientific paper can be of different types, including:

- Scientific articles (Papers)
- Letters
- Reviews

And shorter versions, including:

- Technical notes
- Brief communications
- Research notes
- Perspectives

Each category is appropriate for conveying different types of messages. Generally, scientific articles and letters are valid means for the publication of original research results. Review articles primarily cover the discoveries of other scientists rather than the author's own. This does not mean they are less important. Reviews, fulfill a fundamental function by synthesizing the results of the literature and other sources. Reviews generally highlight misinterpretations or areas of knowledge that require emphasis and future exploration. The structure and format of reviews are less standardized than those of papers or letters.

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Most researchers generally have healthy egos. This means they think their writings deserve the attention of more readers than they actually attract. This difficulty can lead to an inappropriate selection of a potential journal where we would like to publish, and this lack of judgment can lead to delays in publication, excessive manuscript revisions, or an immediate rejection of the publication. Simple but complementary questions that can help select the most appropriate audience are:

- What relevance does this have? This question can be phrased in alternative terms. What impact could my message have on established concepts and/or practices?
- Why should readers pay attention to my message? Could my message lead to a substantial change in the current view of a given subject?
- Who cares? This question could also be expressed in other terms. Who would be most interested in this information? The specialist in a specific discipline? The majority of the community in the area? the scientific world in general? Or the general public?

You must be realistic. It is not a good practice to contemplate a vast potential audience that "needs" to know your information. The more precisely you can answer these questions, the easier it will be to select a journal for publication, and the more precise the target journal, the higher the chances of getting the manuscript published.

Despite the numerous publishing opportunities available in today's digital era, which enable authors to reach a broader audience, selecting a journal for publishing research has become increasingly challenging due to the significant rise in the number of journals in recent years. Even within a specialized area, journals differ in vital aspects such as coverage, format, publication speed, acceptance rate, page costs, and presumably prestige. It is a good practice to use the literature review process to identify potential journals in which we could publish our research work. If you are unsure of the options, start by informing yourself about the journals in your research field. Additional potential journals include those that authors cite in their own research, journals they review for, and journals associated with their professional organizations. Colleagues, the Internet, and, of course, databases are logical sources of information. Compilations of Instructions to Authors (ITAs) published by various journal publishers can help us compare the focus, objectives, and submission requirements. In general, this should be the starting point when evaluating potential journals for publication.

The subsequent step involves examining the publications within the journal you are considering to evaluate the scientific rigor and editorial quality. Key indicators of reputable journals include transparency regarding their aims and scope, the composition of the editorial board, indexing status, peer review process, reputation, and author policies. Assessing these criteria can help identify quality journals that are appropriate for your research publication.

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In general, if a journal publishes a significant number of articles in your area of work, the chances of your article being published are probably higher than if the opposite is true. However, always keep the possibility of publishing in journals outside your field. Nowadays, journal editors accept articles based on their importance to the journal's audience rather than on a specific field of knowledge. Checking databases such as the Science Citation Index can help in making a decision. Table 4, summarizes some specific questions to solve when selecting journals for publication.

Table 4. A few questions to consider when selecting a Journal to publish a research article

What type of journal is it? Is the subject of my paper within the areas of interest of the journal? Are many or few articles on the subject matter of my interest published in the journal? What is the size and type of the publication's audience? What is the rejection rate? What formats are acceptable for the journal? How long does it take to publish a paper in that journal? What is the quality of the graphic reproductions? Are there open access or hybrid options? If using AI assistance in your research process and/or manuscript preparation: What is the policy of the journal regarding AI in manuscript preparation and peer review?

There is no doubt that, for the scientific community, there are journals with different degrees of "prestige." It is not just about the degree of circulation, as many of the most prestigious journals have relatively low circulation volumes. International journals are categorized according to their impact factor and other metrics[31], [32].

Comparing the impact factors of various journals in a particular field of knowledge gives a realistic and quantitative idea of their relative intellectual influence. As of the date of writing this article, the impact factor ranges from 0 to 500+ for the year 2024. It is also worth noting that this index varies annually. However, many detractors of this measurement system argue that the system values the quality of articles rather than the quality of journals. Therefore, the evaluation of impact factors should be understood as indicating the perceived "quality" of a given publication. Table 5 illustrates the typical range of impact factors in different areas of science, based on the latest data from the Journal Citation Report 2024.

Table 5. Typical Impact Factor Ranges by Knowledge Area

Knowledge Area	**Impact Factor**
Biological/Medical Sciences	0-500+
Engineering	0-85+
Physics	0-50+
Chemistry	0-60+

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Mathematics	0-20+
Social Sciences	0-16+

Source: Journal Citation Report 2024

These values reflect the broader range of impact factors observed across various disciplines in recent years, highlighting the increased citation rates and the growing influence of high-impact journals in these fields. As wisely expressed by Schimel [2] on his book on Writing Science, “publish” or “perish” is about surviving in the academic world, not succeeding. You don’t succeed as a scientist by getting papers published. You succeed as a scientist by getting them cited, which serves as a proxy for how useful a research work is.

AI-driven tools can assist researchers in identifying suitable means of publication and appropriate journals for their research. These tools can analyze the content of a manuscript and match it with the scope and focus areas of various journals. By considering factors such as impact factor, acceptance rates, and publication timelines, AI can provide recommendations for journals that align with the research topic. Furthermore, AI can help researchers understand the preferences of different publishers and journals, such as preferred article types and formatting guidelines, ensuring that the submission meets the journal's requirements, thus increasing the chances of acceptance. Nowadays, many publishers already provide AI assisted tools that can help in the process of identifying suitable journals for publication. Providing the title, abstract and potential keywords such tools match your manuscript to the most appropriate scientific journals edited by the publisher. The following are some relevant examples:

- Elsevier Fingerprint Engine™, JournalFinder (https://journalfinder.elsevier.com/)
- Springer Journal Suggester (https://journalsuggester.springer.com/)
- Journal/Author Name Estimator (JANE) (https://jane.biosemantics.org/)
- Edanz Journal Selector (https://www.edanz.com/journal-selector)
- IEEE Publication Recommender™, (https://publication-recommender.ieee.org/home)
- Wiley Journal Finder, (https://journalfinder.wiley.com/search?type=match)

The Think, Check, Submit (https://thinkchecksubmit.org/) service provides further information on what to look out for when assessing a journal and provide easy to follow check lists that would help identify if a given journal or publisher is suitable for your research. These tools can significantly streamline the process of selecting the most appropriate journal for your manuscript by considering various factors such as the scope, impact, and readership of potential journals.

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It is clear that most of us would like to publish in prestigious journals with high circulation and a high impact factor. Every researcher knows which publications in their field meet these criteria, and we would love to build a reputation by publishing in them. However, it is also a fact that high-prestige journals have a rejection rate of up to 90% in some cases. The overall global average in acceptance rates stands around 35-40% [32], [33]. Submitting a paper to this lottery means being in limbo for a very long time (months in some cases) only to receive a rejection. If, after going through all these steps carefully, you still feel unsure about whether you have made an appropriate selection of the journal in which you wish to publish, it is acceptable to write to the editor and ask the necessary questions.

Up to this point, you have taken the initial steps in the process, *i.e.*, planning, collecting, and organizing information. Writing the first draft of the manuscript is the next stage. This stage begins with a series of steps that most people generally do not give due importance to. This includes organizing thoughts through outlines, concept maps, and the development of tables and figures. Once these steps have been taken, the "real" writing of the first draft can begin. It is advisable to write the first draft continuously without stopping much to correct the style; then, it is recommended to set it aside for a few days. AI can assist in providing rough outlines for a first draft but care must be placed in applying best practices in the use of AI-driven tools as outlined in the following section.

Review, an essential stage of the writing process, is the next step. Normally, during the writing of a manuscript, it is common to find things in places where they should not be or even include information that later turns out to be superfluous. One reason for this is that thinking, planning, writing, and reviewing are not separate processes since writing is a tool in the thinking process. We use words as they come to mind; however, our first thoughts are not necessarily the best and may not be organized in the most coherent way. However, given the close link between writing and thinking, when we finish writing, our understanding of the subject has improved. The review process should be thought of as a series of increasingly smaller scale tasks. The first review focuses on organization and logic; the second on more global aspects such as phrasing and style; the third on refining grammar, punctuation, and other related aspects. Ensure you do your due diligence in the language and tone you select. If your research involves working with people with disabilities or issues related to specific groups, the author encourage researchers in those fields to adhere to the basic principles of inclusive language, which are to choose appropriately specific terms and to show respect by calling people what they call themselves in line with APA's (American Psychological Association) Inclusive Language Guide[34].

Writing is only one of the many activities that a researcher must perform on a regular basis. One must know when to end the review process. Apply a benefit-cost principle that applies to almost all human activities including reviewing. This principle states that 20% of the effort is responsible for 80% of the results and that the remaining 80% of the effort generates only 20% of the results. Always keep the

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tasks to be performed in check, *i.e.*, do not lose sight of the forest by focusing on the trees. A to-do list can help us enormously to keep the task in perspective (See Table 6).

Table 6. Main steps in the preparation of a scientific article for publication

1. Collect the necessary information (literature/experimental data/Simulation results) to answer the research question.	12. Review and choose the words that best convey the message for the audience you are addressing.
2. Based on the information available, assess how ready you are to publish the results and in which direction the work is heading.	13. Check grammar, punctuation and other stylistic aspects.
3. Choose a potential journal for publication; obtain the Instructions for Authors from the selected journal and copy sample figures, tables and references. Pay attention to the editorial policies regarding AI usage.	14. Provide a copy to your colleagues for critical review and if possible share the document on a collaborative platform so that each co-author can introduce changes and suggestions.
4. Write a title and choose the sections to include. Its admissible to use AI-bots at this stage to help you generate ideas and potential structure for the manuscript.	15. Get the feedback from your colleagues and revise your manuscript accordingly.
5. Construct concept or mind maps, idea trees or other diagrams to help organize your ideas.	16. Obtain permissions to reproduce previously published material, if necessary.
6. Write the first draft. If LLMs are used in this process, keep records of prompts and answers from the AI bot as you may need to include them as supplemental information.	17. Review the Instructions for Authors, make sure that the final document complies with the format, is a complete document, and contains no spelling errors.
7. Using an appropriate tool, prepare a preliminary reference list.	18. Prepare a final list of references in the appropriate format for the selected journal, ensure that references have been made appropriately within the body of the manuscript and that the required format is used consistently.
8. Put your manuscript aside for a few days to get a better perspective during the revision stage.	19. In collaboration with your colleagues, give some consideration to what kind of publication method is best for your research, open access or a paid journal, if necessary revisit the journal selection stage.
9. Choose and design tables and figures.	20. Submit the manuscript and pay attention to review process and follow up.
10. Review manuscript structure if necessary.	

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It is clear now that generative artificial intelligence (AI) tools, including LLMs are radically altering the way we live and work. Scholarly writing being one of the areas that have significantly been disrupted [8], [14], [15], [35], [36]. However, specific challenges concerning possible misuse of LLMs in science are arising in addition to

ethical risks[37]. If AI bots are to be used in any part of the workflow leading to a manuscript preparation, is time to remind yourself and co-authors on important aspects of best practice when using AI assistance.

3. Ethical Considerations and Best Practices for Using AI in Scientific Writing

While AI tools offer significant benefits in scientific writing, they also pose ethical challenges that must be navigated carefully by authors. The use of AI in scholarly writing requires adherence to ethical norms to maintain academic integrity and trust in the research community.

One primary concern is the potential for plagiarism. AI tools can generate text that closely mimics existing literature, making it difficult to distinguish original work from AI-generated content. Researchers must ensure that all AI-generated content is properly attributed and that the use of AI tools is transparently disclosed in their manuscripts. Failure to do so can lead to accusations of academic dishonesty and damage the credibility of researchers and institutions.

Another ethical consideration is the potential bias in AI algorithms. AI tools are trained on existing datasets, which may contain inherent biases. These biases can be perpetuated in AI-generated content, leading to skewed or unrepresentative findings. Researchers must critically evaluate the output of AI tools and verify that their conclusions are based on robust and unbiased data.

To address these challenges, Table 7 illustrates several best practices that can be adopted when using AI in scientific writing[38]:

Table 7. Best practices when using AI tools

Transparency:	Clearly disclose the use of AI tools in the manuscript, including the specific tools used and the extent of their contribution.
Attribution:	Properly attribute AI-generated content and differentiate it from original research contributions.
Critical Evaluation:	Regularly review and validate AI-generated content to ensure accuracy, relevance, and freedom from bias.
Ethical Training:	Engage in continuous learning about the ethical implications of AI in research and writing, staying informed about best practices and guidelines.

Source: Adapted from [38]

For an illustrative example on Artificial Intelligence (AI)–Assisted Technology usage, the recommendations by the International Committee of Medical Journal Editors (https://icmje.org/) can be used as a starting point. By following these recommendations and best practices, researchers and authors can leverage the

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power of AI to enhance their scientific writing while maintaining high ethical standards and academic integrity.

In addition to plagiarism and potential biases, researchers should be aware of several other risks associated with using AI in scholarly writing[14], [39]. Here are some additional concerns that yourself and co-authors should be aware of:

Issues	**Risk**	**Mitigation measures**
Data Privacy and Security	When using AI tools, researchers often input sensitive data, including unpublished research, personal information, or proprietary information. If the AI platform does not have robust security measures, this data could be at risk of exposure or theft.	Ensure that the AI tools used comply with data protection regulations such as GDPR (General Data Protection Regulation) compliance or HIPAA (Health Insurance Portability and Accountability Act) compliance. Use platforms with strong security protocols and consider anonymizing data before input into any AI-bot.
Misinterpretation of AI Output	AI tools can generate outputs that might be misinterpreted by researchers. For instance, AI might suggest incorrect references or conclusions that seem plausible but are not scientifically sound. Keep in mind that LLMs are very eloquent in generating text but have close to zero critical thinking.	Researchers should critically evaluate AI-generated content, cross-check facts, and avoid relying solely on AI for critical interpretations. Collaborate with peers to review AI outputs.
Over-reliance on AI	There is a danger that researchers might become overly dependent on AI tools, potentially stifling creativity and critical thinking. Over-reliance on AI can lead to a lack of engagement with the research process and a superficial understanding of the subject matter.	Use AI as an aid rather than a replacement for human insight and analysis. Encourage continuous learning and critical thinking skills.
Reproducibility Issues	AI algorithms, especially those that involve deep learning, can be opaque ("black box"), making it difficult to understand how decisions are made. This lack of transparency can lead to challenges in reproducing results.	Choose AI tools that provide transparency and explanations for their outputs. Document AI processes and parameters meticulously to ensure reproducibility.
Ethical and Legal Compliance	Using AI tools might inadvertently lead to breaches of ethical guidelines or legal requirements, especially if the tools are used to generate data or conduct analyses that fall outside standard research	Stay informed about the ethical guidelines and legal requirements relevant to your field. Use AI tools in a manner that complies with these guidelines and seek ethical clearance where necessary.

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	practices.	
Quality of AI Training Data	The quality of AI outputs depends heavily on the data used to train the models as well as on the language used in the training process. If the training data is of poor quality or unrepresentative, the AI outputs will likely be flawed.	Use AI tools trained on high-quality, diverse datasets. If possible perform the prompting in english which is the language more extensively used to train GPTs and LLMs. Understand the provenance and limitations of the training data and apply human judgment to validate AI-generated content[18].

By being aware of these additional risks and taking appropriate measures to mitigate them, researchers can effectively harness the benefits of AI in scholarly writing while maintaining the integrity and quality of their research.

The following are some additional safeguards to keep in mind[40]:

Follow the rules of the target journal. On a regular basis consult and follow an up-to-date version of the rules and policies for the selected journal prior to using an LLM for research or writing assistance. As these tools gain in popularity and its use continue to evolve, journals and publishers are likely to provide explicit guidelines on what they consider to be acceptable or unacceptable uses in submitted manuscripts. It is very important to keep in mind that journal policies are vastly different and what is acceptable for one might not be unacceptable for another.

Understand relevant risks before LLM use. As indicated previously the use of AI tools is not without risks. LLMs use can lead not only to benefits but also to a number of well documented issues, researchers who would like to use AI for their research design and writing endeavors should also first identify relevant risks. Based on an assessment of the risks posed by a particular use case, measures to manage those risks should be adopted such that potential harm is minimized.

Prevent plagiarism. Due to the number of increasing challenges for legal frameworks surrounding copyright and intellectual property that AI presents to the scientific community and the public in general, researchers must proactively seek that the usage of LLMs is safeguarded from plagiarism within its relevant context. Authors also should be advised on the use of AI tools during the peer review process such that this well establish process of constructive criticism might not lead to unfair or superfluous feedback. It is import to keep in mind that until recently literally no attention was paid to the fact that the peer-review process can also be subject to plagiarism[41]. Identified issues in the peer-review system might include, embracing subjectivity, bias, abuse, non-detection of errors, as well as fraud and misconduct by reviewers or journal editors.

Be aware of confidentiality. Unlike standard analytical tools regularly used to prepare manuscripts, sharing confidential data or information with LLMs might

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lead to a potential breach of contract or even more complicated legal issues and must be avoided unless explicit permission is obtained or you have a very clear understanding of the rules established by your institution in regards to AI usage and data confidentiality agreements. This is important to keep in mind because researchers wishing to generate or debug computer code, summarize content, or improve manuscript writing might be tempted to share code, data, published articles or writing with an AI-bot. Although simple in principle, doing so presents many potential problems because it still remains unclear whether GPTs in particular might “remember” this information and then share it with third parties who ask similar questions. In fact at the time of writing this document, OpenAI openly states that conversations established –including data shared-- with its AI-bot are indeed used for further training of the model and warns potential users of this fact. Scientists who work with sensitive, confidential data, for example medical records or defense related research, must keep this in mind.

Always verify the correctness of content generated by an LLM. Outputs *e.g.*, code scripts or content (text) generated by an LLM must always be double-checked by a subject matter expert for factual accuracy. LLMs are well known to “hallucinate”--in other words, to give seemingly correct but false answers-- particularly in response to hard or complex questions[42].

4. The Final Copy: Attention to Detail

It is time to ensure that the "final" manuscript has been written and formatted according to the journal's requirements. Examine a recent issue of the selected journal to check the style. Pay close attention to seemingly insignificant details such as citation style in the text, font type in section and subsection titles, section numbering, use of footnotes, and the design of tables and figures. Remember that no matter how attractive your manuscript pages may be, if they are not in the appropriate format for publication, this can cause complications during the review process. Never under any circumstances attempt to reduce font size or decrease margin sizes to meet the page number requirement of some journals; such an attempt to deceive the editors will only ensure immediate rejection of your manuscript. Read the Instructions to Authors (ITAs) carefully regarding the type and size of font required, use of keywords and indexing codes (PACS). Unless directly approved by the journal, do not send figures or tables reduced in size or quality. It is always advisable to number all pages, including references, tables, figures, etc. Make it your personal policy never to use references you have not personally reviewed, particularly those suggested by a LLM. This can save you future complications from using references or concepts inappropriately.

Finally, have your co-authors or other colleagues read the final document. It is surprising how many small errors can still be detected by individuals who are not part of the process. It may be somewhat embarrassing to find such errors, but it is

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certainly less so than finding them referenced by a referee or, even worse, on a published page.

If using AI bots in the process of manuscript preparation, remind yourself and your co-authors that the output of AI-bots needs to be considered carefully and in line with the journal's current policies on the use of such tools. AI-bots can not be held responsible for inconsistencies, errors in data or citations, or ethical britches. As it stands today, all authors in a manuscript are responsible for its contents.

4.1 Open access vs a paid journal

Today, research is shared through various channels, including blogs, social media platforms, open access journals, and traditional subscription-based journals. Open access journals allow anyone with internet access to freely view their content, but they charge researchers a fee to publish their work. Conversely, paid journals require readers to pay a fee or subscription to access their content. Both types of journals are monitored for impact factors and included in bibliographic databases, the only difference is their business model. While the idea of making journals freely available to the public without financial barriers is appealing, researchers should carefully consider the differences between open access and traditional paid journals before deciding where to publish their research[43], [44]. Most publishers and journals offer detailed information about open access options, so it is important for researchers and their co-authors to fully understand the distinctions between these two publication methods. Some key aspects to consider include: visibility and prestige, accessibility, speed of review process and cost. Open access journals do not make money from subscriptions, so they typically charge higher fees after acceptance.

A viable option that combines the advantages of both open access and traditional publishing is to publish your work in a traditional journal while also making a version available on your personal website or through an open-access repository (such as PubMed Central for biomedical and life sciences, arXiv for mathematics, physics, and computer sciences, medRxiv for health sciences, or chemRxiv for chemistry). Different journals have varying policies on this type of archiving, but most have been compelled to permit it. Some journals allow authors to post the pre-print version of their article online with open access, while others permit the final version to be posted after an embargo period in social scientific networks such as ResearchGate and Academia.

Ultimately, the decision comes down to your personal preference. If you do not feel comfortable entrusting your publication to an open-access journal, then opt for a subscription-based journal publisher that you know and trust.

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4.2 Submission of a Manuscript for Publication

Regardless of how the journal receives the information of your manuscript, some considerations are of interest and facilitate the process. Under no circumstances should you neglect this stage!

Most editors use online electronic submission of manuscripts in various formats, including Word documents (.doc), LaTeX (.tex), and PDF uploaded to dedicated platforms. Electronic submission of manuscripts saves time and money for editors. Moreover, editors can then edit the document and, if necessary, reformat it to meet their printing standards. An additional advantage is that the paper can be quickly put online for access by users subscribed to online bibliographic services or in open access data bases or servers if the authors choose this option for publication. Many journals give priority and, in some cases, exemptions from publication costs for using specific formats for such publications. It is advisable to take advantage of such privileges since it not only saves publication costs but also can significantly accelerate the publication time. Based on the specific requirements of the chosen journal, do not forget to include with your manuscript:

- *Cover Letter*: Write and edit this letter very carefully as this is the first impression the editor has of your work. Make sure to spell the editor's name correctly, check that the address is current and correct. Sending a manuscript to the wrong person can be a mistake that delays publication for some time. In the letter, specify clearly the journal to which you are submitting the manuscript and mention something positive (but not effusive) about why that journal is suitable for publishing your work. Also, mention the title of your manuscript and include your correspondence address as well as your email as the corresponding author.

- *Authorship Form & Statement of Conflicts of Interest declaration*

High rated journals require that each author in a manuscript fulfill certain criteria in the definition of authorship as stated for example by the International Committee of Medical Journal Editors as described below (https://icmje.org/):

Each author is expected to have made substantial contributions to the conception or design of the submitted work;
or the acquisition, analysis, or interpretation of data;
or creation of new software used in the work;
or have drafted the work or substantially revised it;
and has approved the submitted version (and any substantially modified version that involves the author’s contribution to the study;

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and agrees to be personally accountable for the author's own contributions and for ensuring that questions related to the accuracy or integrity of any part of the work, even ones in which the author was not personally involved, are appropriately investigated, resolved, and documented in the literature.
All authors in a manuscript must disclose their own affiliations, funding sources, or financial holdings that might be perceived as sources of bias so that readers can evaluate the data and opinions presented.

- *Copyright Transfer Form*: All authors must sign this form or empower the corresponding author to make this copyright transfer.

- *Data policies and Material sharing:* Authors need to make sure that all data necessary for a reader to understand and evaluate the conclusions of the paper are provided in the manuscript and supplementary information or are archived in an approved public database or repository. The name of the database and accession numbers or link to the repository should be included in the data accessibility statement in the acknowledgment section of the manuscript. If necessary, a statement should be included in the acknowledgment section where the authors affirm that after publication, all reasonable requests for materials will be fulfilled.

- *AI usage statements:* As the issues related with the usage of AI assistance in scientific writing continue to evolve, based on the selected journal you might be required to acknowledge the assistance of AI chat-bots or image generators to prepare your manuscript. Do not include ChatGPT or any AI-bot as co-author of your manuscript.

Once the manuscript is sent, always keep a copy of the manuscript and supplementary material in the exact form it was sent to the editor. Under no circumstances make modifications to this material unless the editor specifies so. Remember Murphy's Laws—if something can go wrong, it will go wrong at the least expected moment. Be particularly careful to appropriately label all the material you have sent and follow the ITAs in this regard.

Now is the time to find something else to do; soon you will receive a communication indicating that your manuscript has been received, but it can take weeks or months before you hear further news about your manuscript.

5. Editorial Review

The receipt of your manuscript by the editor marks the beginning of a series of steps that can culminate in the publication of your manuscript in electronic, printed, or both forms. Only two of these stages involve the authors—the response to comments made by the referees and the review of proofs for printed journals.

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When your manuscript arrives at the editorial office, it is cataloged, dated, and assigned a code that allows it to be tracked through the following stages. The manuscript is reviewed to ensure that all illustrations are included and that it meets the formatting and style requirements of the publication to which it was submitted. The editor then identifies two or three qualified external referees to review your manuscript and sends them copies of your manuscript for evaluation.

Referees are usually asked to send their comments within three weeks of receiving the manuscript. In an ideal world, referees should return their comments promptly, as peer reviewers are usually researchers who are themselves interested in receiving comments on their manuscripts promptly. However, since referees donate their time, some are more conscientious than others about sending their responses within the established time limits. Once the editor has all the responses from the external referees, he or she decides whether or not the paper should be accepted. The manuscript, along with the referees' comments, the editor's decision, and recommendations, are sent to the corresponding author.

If the manuscript is rejected, the author has the option to appeal the decision with the editor or the board of editors. Alternatively, the author(s) can choose to revise the paper, incorporate the recommendations they consider relevant from both the referees and the editor, and resubmit the manuscript to another journal, initiating the publication process anew.

Acceptance of a paper generally also includes some revisions. Be prepared for this. Always take the process with respect. The peer review process is not a perfect process, just a necessary one. The peer review system, where anonymous referees evaluate manuscripts and send comments to authors through the editor, is under no circumstances a pleasant process despite how common it is. Even if you carefully follow the scientific writing process briefly summarized in this article and adopt some of the suggestions, it is always necessary to be prepared for the possibility of receiving negative comments from the referees. The worst thing you can do when receiving negative comments is to treat them as a personal affront to your professional image. On the contrary, calmly and carefully evaluate all the comments received point by point. Make those suggested changes that seem to have more value. Carefully review those that seem incorrect. If a referee misinterprets your message, try to determine why. The misunderstanding may indicate a weakness in the arguments used in the analysis or a part of the document where the writing is weak or unclear. Return the revised manuscript to the editor accompanied by an electronic copy if requested. In your response letter to the editor and referees, address all comments. Indicate acceptance of those suggestions that seem to have merit. If you decide not to accept a comment from the referees, clearly indicate why. Diplomatically and briefly explain your reasoning related to changes you consider unjustified. Avoid responding with anger and any tendency to sarcasm in your responses. Editors send your comments to the referees for whom you may not be anonymous. Remember that the scientific world is really very small, especially when it comes to scientific research.

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If the review of a paper has been extensive, the editor may choose to carry out the peer review process again, essentially returning to the first round. Once the manuscript is fully accepted, the editor sends it to the assemblers who format the manuscript to fit the design and format requirements of a specific journal. Subsequently, the editor sends it to the publishing house. The publishing house reformats (in some cases rewrites) the manuscript and returns what is called a galley proof to the corresponding author for approval. Other procedures may be required at this stage. Generally, the publishing house includes with the galley proofs a form to order reprints of the paper. If there are page charges, the corresponding author will also receive an invoice for the number of pages that have charges, such as pages with color figures.

Proofreading is, still in the era of technology, an important stage in the publication process. Take it seriously and give it the importance it deserves. With the use of pre-established formats, fewer errors are introduced during the reformatting process, but finding error-free proofs is unlikely. Examine each paragraph individually, word by word, and line by line. Compare the original material of the accepted manuscript with the proof material and ensure that no material has been omitted or repeated. Carefully examine numbers in tables, figures, and equations. Errors at this stage are very difficult to detect if working alone, so it is advisable for several people to read the proofs. Although proofreaders' marks are now used much less often than in the past, when proofreading a document, you must understand what each proofreading mark means to know how best to correct it. It is very important to familiarize yourself with the international symbols used for proofreading. Several websites can be scanned for common proofreader's markings. Mobile applications as well as professional service providers are also now available that can simplify the process at a cost. The most common corrections are additions, deletions, changes in capitalization, word order, punctuation, or font type. Mark corrections in the margin of the proof and insert symbols indicating the type of change to be made within the text of the document where the error occurs. Never write corrections above the line. If necessary, use an additional sheet to describe the required correction in detail. Mark each change in the proof. Proof read your manuscript more than once.

Proofs are very sensitive to correction time from the publishing house's perspective. Read and correct the proofs as soon as possible and return them to the publishing house. Delays at this point are very costly for all entities involved. The time for review and addition of new information has passed. At this point, only the correction of errors is allowed. Other changes are expressly prohibited.

5.1 AI in Peer Review

Just as AI tools has significantly enhance the efficiency and accuracy of the manuscript preparation and submission process described above; AI has also

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made its way into the peer-review process with very positive impacts for the peer-review system but also with emergent issues such as the noted phenomenon of peer-review plagiarism[41], [45]. An emergent modality of plagiarism in which anonymous reviewers simply copy and paste or paraphrase reviews from multiple web sources --particularly journals sharing open reviews as a part of their service-- or generate the reviews to a manuscript by means of AI tools. More worrisome for the scientific community is that peer-review plagiarism has been generated not only by referees but also by editors of predatory journals with the aim to hide the actual absence of a rigorous peer-review process. This problem has been apparently gaining in strength in recent years and authors should be aware of such malicious practices which can negatively affect many in the scientific community.

Recently, based on the several identified shortcomings of the human dependent scientific review process, arguments have been made for the incorporation of new AI-driven technologies to assess scientific merit and novelty, thus enabling AI-dependent review of submitted articles[46]. However, many questions and concerns are emerging as to how effective these AI bots are, if we must rely on their reviews without any human input or intervention[47], [48], [49].

6. Issues after publication

Mistakes can occur in published journal articles, and some require the publisher to not only amend the article but also issue a correction notice. This notice is a formal, public declaration that informs readers of the changes made to the published work. Such corrections may be referred to as errata or corrigenda. If you discover an error in your published article (including an online first article), the initial step is to notify the journal's editor and publisher about the mistake. They will assess whether a formal correction notice is necessary. This notice is important to update the knowledge base for both current and future readers of the article. Should a correction notice be deemed necessary, you will be responsible for writing it. When communicating with the journal editor, provide a proposed correction notice that details the nature of the error, the correct information, and whether all versions of the original article have been corrected.

Once the correction notice is approved, it is formatted using the journal's official template with the information supplied by the author. This correction notice is typically assigned a DOI (*Digital Object Identifier*) and published both in print and online. Additionally, the correction notice is linked to the article's record in research databases, ensuring that readers will see the notice when they access the article or its database entry.

In some cases, the identified errors in an article are so significant that retraction becomes necessary. Circumstances that may lead to a retraction include:

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- A major unintentional miscalculation or error that invalidates the study's findings or conclusions.
- Falsification or fabrication of data (including images) or other forms of misconduct.
- Significant plagiarism or duplicate publication, along with copyright infringement.
- Unethical research practices used in the study.
- The identification of a major undisclosed competing interest that could have influenced the results or their interpretation.

When articles are retracted, they are not removed from the publisher's databases. Instead, they remain accessible but are clearly marked with a retraction notice, and bibliographic databases are informed. Retractions are increasingly common and can often result in severe career consequences[50]. The likelihood of future scandals involving retractions due to author misconduct is high, highlighting the need for students, young researchers, and scientists to be well-prepared to avoid such situations.

OUTLOOK

Scientific writing is a critical component of the research process, ensuring that discoveries are communicated effectively and preserved for future reference. This article outlines the fundamental steps involved in writing and publishing a scientific paper, emphasizing the importance of clarity, organization, and adherence to ethical standards.

Starting with the identification of a research question, the process involves meticulous planning, data collection, and literature review. The role of AI in enhancing these stages has been highlighted, indicating how AI tools can assist in drafting, proofreading, and organizing manuscripts. AI can also help in identifying suitable journals for publication and streamlining the submission process, thereby improving the efficiency and quality of scientific writing.

However, the integration of AI in scholarly writing is not without challenges. Ethical considerations such as avoiding plagiarism, ensuring data privacy, and mitigating biases in AI outputs are paramount. Researchers must use AI responsibly, maintaining transparency about the use of these tools and critically evaluating AI-generated content to uphold the integrity of their work.

The manuscript preparation phase requires careful attention to detail, from formatting according to journal guidelines to addressing reviewer comments constructively. AI can aid in these tasks, but researchers must remain vigilant to ensure that AI contributions are accurate and ethically sound.

The editorial review process, including peer review and proof correction, is essential for maintaining the quality of published research. AI tools can facilitate

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this process by suggesting appropriate reviewers, managing submission materials, and even identifying potential ethical issues in manuscripts but emergent misconduct practices in some cases by referees and in others by editors need to be addressed urgently by all those involved (authors, editors and publishers).

In conclusion, while AI offers significant advantages in scientific writing, researchers must balance its use with ethical considerations and critical oversight. By doing so, they can enhance the clarity, accuracy, and impact of their publications, contributing to the advancement of knowledge and the integrity of the scientific community. The guidelines and best practices briefly discussed in this article provide a basic framework for effectively integrating AI into the scholarly writing process while maintaining high standards of academic integrity.

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REFERENCES

[1] J. R. Matthews and R. W. Matthews, *Successful scientific writing: a step-by-step guide for the biological and medical sciences*, 3rd ed. Cambridge ; New York: Cambridge University Press, 2008.

[2] J. Schimel, *Writing science: how to write papers that get cited and proposals that get funded*. Oxford ; New York: Oxford University Press, 2012.

[3] J. K. Iskander, S. B. Wolicki, R. T. Leeb, and P. Z. Siegel, "Successful Scientific Writing and Publishing: A Step-by-Step Approach," *Prev. Chronic. Dis.*, vol. 15, p. 180085, Jun. 2018, doi: 10.5888/pcd15.180085.

[4] M. E. Emenike and B. U. Emenike, "Was This Title Generated by ChatGPT? Considerations for Artificial Intelligence Text-Generation Software Programs for Chemists and Chemistry Educators," *J. Chem. Educ.*, vol. 100, no. 4, pp. 1413–1418, Apr. 2023, doi: 10.1021/acs.jchemed.3c00063.

[5] M. Khalifa and M. Albadawy, "Using artificial intelligence in academic writing and research: An essential productivity tool," *Comput. Methods Programs Biomed. Update*, vol. 5, p. 100145, 2024, doi: 10.1016/j.cmpbup.2024.100145.

[6] M. S. Schäfer, "The Notorious GPT: science communication in the age of artificial intelligence," *J. Sci. Commun.*, vol. 22, no. 02, May 2023, doi: 10.22323/2.22020402.

[7] G. Grimaldi and B. Ehrler, "AI *et al.* : Machines Are About to Change Scientific Publishing Forever," *ACS Energy Lett.*, vol. 8, no. 1, pp. 878–880, Jan. 2023, doi: 10.1021/acsenergylett.2c02828.

[8] B. D. Lund, T. Wang, N. R. Mannuru, B. Nie, S. Shimray, and Z. Wang, "CHATGPT and a new academic reality: ARTIFICIAL INTELLIGENCE-WRITTEN research papers and the ethics of the large language models in scholarly publishing," *J. Assoc. Inf. Sci. Technol.*, vol. 74, no. 5, pp. 570–581, May 2023, doi: 10.1002/asi.24750.

[9] C. Stokel-Walker, "ChatGPT listed as author on research papers: many scientists disapprove," *Nature*, vol. 613, no. 7945, pp. 620–621, Jan. 2023, doi: 10.1038/d41586-023-00107-z.

[10] J. Gonzalo Muga, "Editorial —Artificial Intelligence language models in scientific writing," *Europhys. Lett.*, vol. 143, no. 2, p. 20000, Jul. 2023, doi: 10.1209/0295-5075/ace3ef.

[11] A. Birhane, A. Kasirzadeh, D. Leslie, and S. Wachter, "Science in the age of large language models," *Nat. Rev. Phys.*, vol. 5, no. 5, pp. 277–280, Apr. 2023, doi: 10.1038/s42254-023-00581-4.

[12] T. Teubner, C. M. Flath, C. Weinhardt, W. Van Der Aalst, and O. Hinz, "Welcome to the Era of ChatGPT et al.: The Prospects of Large Language Models," *Bus. Inf. Syst. Eng.*, vol. 65, no. 2, pp. 95–101, Apr. 2023, doi: 10.1007/s12599-023-00795-x.

[13] S. M. Pressman, S. Borna, C. A. Gomez-Cabello, S. A. Haider, C. Haider, and A. J. Forte, “AI and Ethics: A Systematic Review of the Ethical Considerations of Large Language Model Use in Surgery Research,” *Healthcare*, vol. 12, no. 8, p. 825, Apr. 2024, doi: 10.3390/healthcare12080825.

[14] B. Tomlinson, A. W. Torrance, and R. W. Black, “ChatGPT and Works Scholarly: Best Practices and Legal Pitfalls in Writing with AI,” May 04, 2023, *arXiv*: arXiv:2305.03722. Accessed: Jun. 07, 2024. [Online]. Available: http://arxiv.org/abs/2305.03722

[15] P. P. Ray, “ChatGPT: A comprehensive review on background, applications, key challenges, bias, ethics, limitations and future scope,” *Internet Things Cyber-Phys. Syst.*, vol. 3, pp. 121–154, 2023, doi: 10.1016/j.iotcps.2023.04.003.

[16] J. M. Buriak *et al.*, “Best Practices for Using AI When Writing Scientific Manuscripts: Caution, Care, and Consideration: Creative Science Depends on It,” *ACS Nano*, vol. 17, no. 5, pp. 4091–4093, Mar. 2023, doi: 10.1021/acsnano.3c01544.

[17] M. D. Wilkinson *et al.*, “The FAIR Guiding Principles for scientific data management and stewardship,” *Sci. Data*, vol. 3, no. 1, p. 160018, Mar. 2016, doi: 10.1038/sdata.2016.18.

[18] S. Longpre *et al.*, “A large-scale audit of dataset licensing and attribution in AI,” *Nat. Mach. Intell.*, vol. 6, no. 8, pp. 975–987, Aug. 2024, doi: 10.1038/s42256-024-00878-8.

[19] M. A. Hanson, P. G. Barreiro, P. Crosetto, and D. Brockington, “The strain on scientific publishing,” Sep. 27, 2023, *arXiv*: arXiv:2309.15884. doi: 10.48550/arXiv.2309.15884.

[20] S. J. Porter and L. D. McIntosh, “Identifying Fabricated Networks within Authorship-for-Sale Enterprises,” Jan. 08, 2024, *arXiv*: arXiv:2401.04022. Accessed: Jun. 07, 2024. [Online]. Available: http://arxiv.org/abs/2401.04022

[21] G. Cabanac, “Decontamination of the scientific literature,” report, Institut Universitaire de France, 2022. doi: 10.48550/arXiv.2210.15912.

[22] S. Kojaku, G. Livan, and N. Masuda, “Detecting anomalous citation groups in journal networks,” *Sci. Rep.*, vol. 11, no. 1, p. 14524, Jul. 2021, doi: 10.1038/s41598-021-93572-3.

[23] I. Fister, I. Fister, and M. Perc, “Toward the Discovery of Citation Cartels in Citation Networks,” *Front. Phys.*, vol. 4, Dec. 2016, doi: 10.3389/fphy.2016.00049.

[24] G. Cabanac, C. Labbé, and A. Magazinov, “Tortured phrases: A dubious writing style emerging in science. Evidence of critical issues affecting established journals,” Jul. 12, 2021, *arXiv*: arXiv:2107.06751. Accessed: Jun. 07, 2024. [Online]. Available: http://arxiv.org/abs/2107.06751

[25] J. A. Teixeira Da Silva, “‘Tortured phrases’ in preprints,” *Curr. Med. Res. Opin.*, vol. 39, no. 5, pp. 785–787, May 2023, doi: 10.1080/03007995.2023.2201098.

[26] M. Li and Z. Shen, “Science map of academic misconduct,” *The Innovation*, vol. 5, no. 2, p. 100593, Mar. 2024, doi: 10.1016/j.xinn.2024.100593.

[27] R. Van Noorden, “More than 10,000 research papers were retracted in 2023 — a new record,” *Nature*, vol. 624, no. 7992, pp. 479–481, Dec. 2023, doi: 10.1038/d41586-023-03974-8.

[28] B. A. Nosek *et al.*, “Replicability, Robustness, and Reproducibility in Psychological Science,” *Annu. Rev. Psychol.*, vol. 73, no. 1, pp. 719–748, Jan. 2022, doi: 10.1146/annurev-psych-020821-114157.

[29] C. Stoddart, “Is there a reproducibility crisis in science?,” *Nature*, pp. d41586-019-00067–3, May 2016, doi: 10.1038/d41586-019-00067-3.

[30] Committee on Reproducibility and Replicability in Science *et al.*, *Reproducibility and Replicability in Science*. Washington, D.C.: National Academies Press, 2019, p. 25303. doi: 10.17226/25303.

[31] V. Phillips and E. Barker, “Writing for publication: Structure, form, content, and journal selection,” *J. Perioper. Pract.*, vol. 31, no. 6, pp. 230–233, Jun. 2021, doi: 10.1177/1750458921996249.

[32] B. J. Runde, “Time to publish? Turnaround times, acceptance rates, and impact factors of journals in fisheries science,” *PLOS ONE*, vol. 16, no. 9, p. e0257841, Sep. 2021, doi: 10.1371/journal.pone.0257841.

Unpublished DRAFT

[33] B.-C. Bjórk, “Acceptance rates of scholarly peer-reviewed journals: A literature survey,” *El Prof. Inf.*, vol. 28, no. 4, Jul. 2019, doi: 10.3145/epi.2019.jul.07.
[34] “Introducing the second edition of the APA Inclusive Language Guide,” https://apastyle.apa.org. Accessed: Jun. 07, 2024. [Online]. Available: https://apastyle.apa.org/blog/inclusive-language-guide-second-edition
[35] A. BaHammam, “Balancing Innovation and Integrity: The Role of AI in Research and Scientific Writing,” *Nat. Sci. Sleep*, vol. Volume 15, pp. 1153–1156, Dec. 2023, doi: 10.2147/NSS.S455765.
[36] W. Liang *et al.*, “Mapping the Increasing Use of LLMs in Scientific Papers,” Apr. 01, 2024, *arXiv*: arXiv:2404.01268. Accessed: Jun. 07, 2024. [Online]. Available: http://arxiv.org/abs/2404.01268
[37] D. B. Resnik and M. Hosseini, “The ethics of using artificial intelligence in scientific research: new guidance needed for a new tool,” *AI Ethics*, May 2024, doi: 10.1007/s43681-024-00493-8.
[38] B. N. Hryciw, A. J. E. Seely, and K. Kyeremanteng, “Guiding principles and proposed classification system for the responsible adoption of artificial intelligence in scientific writing in medicine,” *Front. Artif. Intell.*, vol. 6, p. 1283353, Nov. 2023, doi: 10.3389/frai.2023.1283353.
[39] J. He *et al.*, “Control Risk for Potential Misuse of Artificial Intelligence in Science,” Dec. 11, 2023, *arXiv*: arXiv:2312.06632. Accessed: Jun. 10, 2024. [Online]. Available: http://arxiv.org/abs/2312.06632
[40] G. R. Smith *et al.*, “Ten simple rules for using large language models in science, version 1.0,” *PLOS Comput. Biol.*, vol. 20, no. 1, p. e1011767, Jan. 2024, doi: 10.1371/journal.pcbi.1011767.
[41] M. Piniewski, I. Jarić, D. Koutsoyiannis, and Z. W. Kundzewicz, “Emerging plagiarism in peer-review evaluation reports: a tip of the iceberg?,” *Scientometrics*, vol. 129, no. 4, pp. 2489–2498, Apr. 2024, doi: 10.1007/s11192-024-04960-1.
[42] H. Shakil, Z. Ortiz, and G. C. Forbes, “Utilizing GPT to Enhance Text Summarization: A Strategy to Minimize Hallucinations,” May 07, 2024, *arXiv*: arXiv:2405.04039. Accessed: Jun. 10, 2024. [Online]. Available: http://arxiv.org/abs/2405.04039
[43] R. G. Dudley, “The Changing Landscape of Open Access Publishing: Can Open Access Publishing Make the Scholarly World More Equitable and Productive?,” *J. Librariansh. Sch. Commun.*, vol. 9, no. 1, Feb. 2021, doi: 10.7710/2162-3309.2345.
[44] L. Stich, M. Spann, and K. M. Schmidt, “Paying for open access,” *J. Econ. Behav. Organ.*, vol. 200, pp. 273–286, Aug. 2022, doi: 10.1016/j.jebo.2022.05.023.
[45] W. Liang *et al.*, “Monitoring AI-Modified Content at Scale: A Case Study on the Impact of ChatGPT on AI Conference Peer Reviews,” Mar. 11, 2024, *arXiv*: arXiv:2403.07183. Accessed: Jun. 07, 2024. [Online]. Available: http://arxiv.org/abs/2403.07183
[46] B. Doskaliuk, O. Zimba, M. Yessirkepov, I. Klishch, and R. Yatsyshyn, “Artificial Intelligence in Peer Review: Enhancing Efficiency While Preserving Integrity,” *J. Korean Med. Sci.*, vol. 40, no. 7, p. e92, 2025, doi: 10.3346/jkms.2025.40.e92.
[47] J. M. Buriak, M. C. Hersam, and P. V. Kamat, “Can ChatGPT and Other AI Bots Serve as Peer Reviewers?,” *ACS Energy Lett.*, vol. 9, no. 1, pp. 191–192, Jan. 2024, doi: 10.1021/acsenergylett.3c02586.
[48] V. Mollaki, “Death of a reviewer or death of peer review integrity? the challenges of using AI tools in peer reviewing and the need to go beyond publishing policies,” *Res. Ethics*, vol. 20, no. 2, pp. 239–250, Apr. 2024, doi: 10.1177/17470161231224552.
[49] M. Naddaf, “AI is transforming peer review — and many scientists are worried,” *Nature*, vol. 639, no. 8056, pp. 852–854, Mar. 2025, doi: 10.1038/d41586-025-00894-7.
[50] P. Azoulay, A. Bonatti, and J. L. Krieger, “The career effects of scandal: Evidence from scientific retractions,” *Res. Policy*, vol. 46, no. 9, pp. 1552–1569, Nov. 2017, doi: 10.1016/j.respol.2017.07.003.

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